

A =LU Factorization

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1. $A = LU$ Factorization

We first point out that the Elimination is equally to the $A = LU$ Factorization. Below is the summary to this Chapter's contents :

- Each elimination step is inverted by L_{ij} . Off the main diagonal change $-e_{ij}$ to e_{ij} .
- The whole forward elimination process (with no row exchanges) is inverted by L .
- The Original matrix A is recovered from U by $A = LU = (\text{Lower triangular})(\text{Upper triangular})$.

1.1. The forward Elimination $EA = U$.

Suppose we have a matrix :

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix}$$

we need to execute the elimination step by step :

- **Step 1** : We first eliminate the second pivot $a_{(2,1)} = 1$:

we need a elimination matrix

$$E_{(2,1)} = \begin{bmatrix} 1 & 0 & 0 \\ -\frac{1}{2} & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

then, multiply it by matrix A to produce :

$$E_{(2,1)}A = \begin{bmatrix} 1 & 0 & 0 \\ -\frac{1}{2} & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix} = \begin{bmatrix} 2 & 1 & 0 \\ 0 & \frac{3}{2} & 1 \\ 0 & 1 & 2 \end{bmatrix} = U_1$$

- **Step 2** : We second need to eliminate the third pivot $a_{(3,2)} = 1$:

$$E_{(3,2)} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -\frac{2}{3} & 1 \end{bmatrix}$$

then similarly :

$$E_{(3,2)}U_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -\frac{2}{3} & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 & 0 \\ 0 & \frac{3}{2} & 1 \\ 0 & 1 & 2 \end{bmatrix} = \begin{bmatrix} 2 & 1 & 0 \\ 0 & \frac{3}{2} & 1 \\ 0 & 0 & \frac{4}{3} \end{bmatrix} = U$$

- **Step 3** : We can multiply the $E_{(2,1)}$ by $E_{(3,2)}$ to produce the Elimination Matrix

$$E = E_{(3,2)}E_{(2,1)} = \begin{bmatrix} 1 & 0 & 0 \\ -\frac{1}{2} & 1 & 0 \\ 0 & -\frac{2}{3} & 1 \end{bmatrix}$$

. thus, we can say :

$$EA = \begin{bmatrix} 1 & 0 & 0 \\ -\frac{1}{2} & 1 & 0 \\ 0 & -\frac{2}{3} & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix} = \begin{bmatrix} 2 & 1 & 0 \\ 0 & \frac{3}{2} & 1 \\ 0 & 0 & \frac{4}{3} \end{bmatrix} = U$$

> where the row 3 of the matrix U come from ?

$$\text{row 3 of } U = (\text{row 3 of } A) - e_{(2,1)}(\text{row 1 of } U) - e_{(3,2)}(\text{row 2 of } U)$$

with the

$$e_{(2,1)} = \frac{1}{2}; e_{(3,2)} = \frac{2}{3}$$

1.2. The backward recover : $A = LU$

Now, we wanna to recover the matrix A from the upper matrix U : the first thing is we need to get the inverse of the Elimination matrix E :

$$\bullet \quad E^{-1}E = I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = E^{-1} \begin{bmatrix} 1 & 0 & 0 \\ -\frac{1}{2} & 1 & 0 \\ 0 & -\frac{2}{3} & 1 \end{bmatrix} \Rightarrow E^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ \frac{1}{2} & 1 & 0 \\ 0 & \frac{2}{3} & 1 \end{bmatrix}$$

so, if we multiply the matrix EA and U by matrix E^{-1} at the same time :

$$\begin{aligned} E^{-1}EA &= E^{-1}U \\ IA &= E^{-1}U \\ A &= E^{-1}U \end{aligned}$$

so the finally we produce the equation :

$$A = E^{-1}U$$

simultaneously, we find out the matrix E^{-1} is a lower triangular matrix, so we denote this as L . thus,

$$A = E^{-1}U = LU$$

$$L = E^{-1}$$

this process is called “ $A = LU$ Factorization”. “Whenever E does, L will undo them.”

1.3. The $A = LDU$ Factorization

we have seen the $A = LU$ Factorization in above section, but the $A = LU$ is not symmetric, we will introduce the symmetric form :

$$A = LDU$$

- with D is a diagonal matrix, the pivots are on its diagonal.
- with L still is the lower triangular matrix.
- the U is upper matrix, which the 1's on the diagonal.

For example :

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix}; U = \begin{bmatrix} 2 & 1 & 0 \\ 0 & \frac{3}{2} & 1 \\ 0 & 0 & \frac{4}{3} \end{bmatrix}; L = \begin{bmatrix} 1 & 0 & 0 \\ \frac{1}{2} & 1 & 0 \\ 0 & \frac{2}{3} & 1 \end{bmatrix}$$

$$A = LU = LDU$$

$\Rightarrow$

$$\begin{aligned} & \begin{bmatrix} 1 & 0 & 0 \\ \frac{1}{2} & 1 & 0 \\ 0 & \frac{2}{3} & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 & 0 \\ 0 & \frac{3}{2} & 0 \\ 0 & 0 & \frac{4}{3} \end{bmatrix} \begin{bmatrix} \frac{2}{2} & \frac{1}{2} & 0 \\ 0 & \frac{\frac{3}{2}}{\frac{3}{2}} & \frac{1}{\frac{3}{2}} \\ 0 & 0 & \frac{\frac{4}{3}}{\frac{4}{3}} \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 & 0 \\ \frac{1}{2} & 1 & 0 \\ 0 & \frac{2}{3} & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 & 0 \\ 0 & \frac{3}{2} & 0 \\ 0 & 0 & \frac{4}{3} \end{bmatrix} \begin{bmatrix} 1 & \frac{1}{2} & 0 \\ 0 & 1 & \frac{2}{3} \\ 0 & 0 & 1 \end{bmatrix} = LDU \end{aligned}$$

Genrally, we can split U into :

$$\begin{pmatrix} d_1 & 0 & 0 & \dots \\ 0 & d_2 & 0 & \dots \\ 0 & 0 & \vdots & \vdots \\ \dots & \dots & \dots & d_n \end{pmatrix} \begin{pmatrix} 1 & \frac{u_{12}}{d_1} & \frac{u_{13}}{d_1} & \dots \\ 0 & 1 & \frac{u_{23}}{d_2} & \dots \\ 0 & 0 & \vdots & \vdots \\ 0 & 0 & 0 & 1 \end{pmatrix} = DU$$

thus, As we need to divide the pivot by values which on each row, so pivots cannot be zero!!!

1.4. Review for above subsection

$$EA = (E_{(3,2)}E_{(2,1)})A = U$$

$$A = (E_{(2,1)}^{-1}E_{(3,2)}^{-1})U = LU = LDU$$